

GEMS TAIWAN iGEM
High School Lesson Plan

Title/Subject: Decoding the Genetic Code: From DNA to the Future of Food Security

Target Group: Grade 10 to Grade 12 students

Time: 1 hour

Learning Objectives:

- **Integrate interdisciplinary knowledge:** Students will understand DNA structure and the central dogma, and connect synthetic biology to real-world applications in agriculture, environmental sustainability, and climate change adaptation (e.g., heat, drought and salinity stress).
- **Develop scientific inquiry and practical skills:** Through an osmosis experiment that simulates salinity stress responses, students will observe, analyze, and infer how environmental stress affects crop growth and food security.
- **Foster innovation and ethical reasoning:** By participating in a mutation simulation game, students will evaluate the advantages and disadvantages of gene editing while developing innovative, science-based solutions to global challenges aligned with the Sustainable Development Goals (SDGs).

Lesson flows:

1. Introduction (10 minutes)

- Kahoot: assess students' familiarity with DNA, synthetic biology and plant stress
- Introduce the iGEM competition
- Provide an overview of synthetic biology and its impact and applications in everyday life

2. Plant Protectants and Biotic & Abiotic Stress and Hands-on Osmosis Experiment (20 minutes)

- **Guiding question:**
 - *What happens to plants as the Earth becomes hotter and drier?*
 - *How does environmental stress affect plant growth, and what are the consequences for food production?*
- Do an experiment to simulate how plants respond to environmental stress.
- Explain the causes of biotic and abiotic stress and their effects on plant growth using simulation
- Shows experiment video of how heat stress affects photosynthesis.
- Introduce our project: using synthetic biology to help plants live under environmental stress
- Explain plant protectants and how they work
- Upload experimental pictures and debrief of experimental result

3. Revision of DNA and mutation (10 minutes)

- Review the structure of DNA and the rules of pairing
- Review the different types of mutation
- Discuss the advantages and disadvantages of genetic mutations.
- How scientists design mutants to control the gene expression of microorganisms to conduct specific tasks

4. Mutation Game (10 minutes)

- Interactive transcription and translation simulation game: help students understand how mutations affect gene expression
 - Students will be lined up and divided into sighter, messenger, and builder
 - Sighter(1): the only person who can look at the given codon. The sighter needs to transcribe DNA to RNA and pass it down to the messenger
 - Messengers(5): pass down the RNA codon based on what they heard from the last messenger
 - Builder(1): based on the codon told by the messenger, translate the RNA codon to the correct sequence of amino acid
 - Through memorizing, passing down the codon, doing transcription and translation, students will experience how mutation happens.

5. Course Summary (10 minutes)

- Review key concepts (including synthetic biology, plant protectants, and DNA).
- Collect students' feedback through a post-course survey.